

DKTE College Training

Self Joins

```
SELECT s1.sname, s2.sname, s1.city FROM
salespeople s1
INNER JOIN salespeople s2 ON s1.city = s2.city
WHERE s1.snum < s2.snum;
```

- <https://leetcode.com/problems/employees-earning-more-than-their-managers/>

Analytical Queries -- Window Functions

```
SELECT ename, job, sal, deptno FROM emp;

SELECT deptno, SUM(sal) FROM emp
GROUP BY deptno;

SELECT ename, job, deptno, SUM(sal) FROM emp
GROUP BY deptno;
-- when group by clause used, only grouped columns can be selected with aggregate
operations on other columns.
```

- GROUP BY Limitation: When GROUP BY is done, you lose details from other columns.
- **Window Functions**
 1. Aggregate functions -- e.g. SUM(), AVG(), MAX(), MIN(), COUNT(), ...
 2. Non-Aggregate functions -- e.g. RANK(), ROW_NUMBER(), DENSE_RANK(), LEAD(), LAG(), FIRST(), LAST(), ...
- **Window**
 - Sub-set of records based on some criteria and may have some order.
 - Defined using OVER() clause
 - OVER() -- all rows in window -- all emps
 - OVER(PARTITION BY colname) -- all rows matching same column value e.g. PARTITION BY deptno => all emps in dept 10, all emps in dept 20, all emps in dept 30
 - OVER(ORDER BY colname) -- all rows in a window sorted by given column name e.g. ORDER BY sal => all emps in asc order of sal
 - OVER(PARTITION BY col1name ORDER BY col2name) -- all rows matching same column(1) value sorted by given column(2) e.g. PARTITION BY deptno ORDER BY sal => all emps in dept 10 sorted by sal in asc order, all emps in dept 20 sorted by sal in asc order, all emps in dept 30 sorted by sal in asc order

```
SELECT ename, job, deptno, SUM(sal) OVER() total FROM emp;

SELECT ename, job, deptno, SUM(sal) OVER(PARTITION BY deptno) total FROM emp;
```

- compare emp sal with avg sal of his job profile.

```
SELECT ename, job, sal, AVG(sal) OVER(PARTITION BY job) avgsal FROM emp;

SELECT ename, job, sal, AVG(sal) OVER(PARTITION BY job) avgsal, sal - AVG(sal)
OVER(PARTITION BY job) diff FROM emp;
```

- how much sal rise should be given to each emp so that his sal is same as max sal for his job profile.

```
SELECT ename, job, sal, MAX(sal) OVER(PARTITION BY job) avgsal, MAX(sal)
OVER(PARTITION BY job) - sal diff FROM emp;
```

- **ROW_NUMBER(), RANK(), DENSE_RANK()**

- ROW_NUMBER() -- row num (sr)
- RANK() -- rank based on given order -- if two rows have same value, then their rank is same. Few ranks may be unused.
- DENSE_RANK() -- same as rank, no number is left unused in between.

```
SELECT ename, sal, deptno, ROW_NUMBER() OVER() sr, RANK() OVER() rnk, DENSE_RANK()
OVER() drnk FROM emp;
```

```
SELECT ename, sal, deptno,
ROW_NUMBER() OVER(ORDER BY sal) sr,
RANK() OVER(ORDER BY sal) rnk,
DENSE_RANK() OVER(ORDER BY sal) drnk
FROM emp;
```

```
SELECT ename, sal, deptno,
ROW_NUMBER() OVER(ORDER BY sal DESC) sr,
RANK() OVER(ORDER BY sal DESC) rnk,
DENSE_RANK() OVER(ORDER BY sal DESC) drnk
FROM emp;
```

- **Window Specification**

- WINDOW wnd AS (ORDER BY sal DESC);
- Changing Window spec at one place will reflect at all its occurrence in that query.

```
SELECT ename, sal, deptno,
ROW_NUMBER() OVER(wnd) sr,
```

```
RANK() OVER(wnd) rnk,
DENSE_RANK() OVER(wnd) drnk
FROM emp
WINDOW wnd AS (ORDER BY sal DESC);
```

- Deptwise salary rankings

```
SELECT ename, sal, deptno,
ROW_NUMBER() OVER(wnd) sr,
RANK() OVER(wnd) rnk,
DENSE_RANK() OVER(wnd) drnk
FROM emp
WINDOW wnd AS (PARTITION BY deptno ORDER BY sal DESC);
```

- Multiple Window Specifications -- e.g. Salwise rank and Namewise rank

```
SELECT ename, sal,
DENSE_RANK() OVER(wnd1) saldrnk,
DENSE_RANK() OVER(wnd2) namedrnk
FROM emp
WINDOW wnd1 AS (ORDER BY sal DESC),
wnd2 AS (ORDER BY ename ASC);
```

- Find of range of salaries.

```
SELECT MAX(sal) - MIN(sal) AS rng FROM emp;

SELECT deptno, MAX(sal) - MIN(sal) AS rng FROM emp
GROUP BY deptno;
```

- First and Last in Partition

```
SELECT FIRST_VALUE(ename) OVER(w) f,
LAST_VALUE(ename) OVER(w) l,
ename, sal, deptno FROM emp
WINDOW w AS (PARTITION BY deptno);
```

- Next and Prev Row -- LEAD() and LAG()

```
SELECT ename, sal,
LEAD(sal) OVER(w) nextsal,
LAG(sal) OVER(w) prevsal
FROM emp
```

```
WINDOW w AS (ORDER BY sal DESC);

SELECT deptno, ename, sal,
LEAD(sal) OVER(w) nextsal,
LAG(sal) OVER(w) prevsal
FROM emp
WINDOW w AS (PARTITION BY deptno ORDER BY sal DESC);
```

- Find difference between sal of emp with sal of next emp (with higher/equal sal) in same dept.

```
SELECT ename, deptno, sal,
LEAD(sal) OVER(w) - sal AS diff
FROM emp
WINDOW w AS (PARTITION BY deptno ORDER BY sal ASC);
```

Derived Tables

```
-- categories emp as follows
-- < 1500 -- POOR
-- BETN 1500 AND 2500 -- MID
-- > 2500 -- RICH

SELECT ename, sal, CASE
WHEN sal < 1500 THEN 'POOR'
WHEN sal BETWEEN 1500 AND 2500 THEN 'MID'
ELSE 'RICH'
END cat
FROM emp;

-- count num of emps in each category
CREATE VIEW emp_categories AS
SELECT ename, sal, CASE
WHEN sal < 1500 THEN 'POOR'
WHEN sal BETWEEN 1500 AND 2500 THEN 'MID'
ELSE 'RICH'
END cat
FROM emp;

SELECT * FROM emp_categories;

SELECT cat, COUNT(ename) cnt
FROM emp_categories
GROUP BY cat;

-- Inline View or Derived Table
SELECT cat, COUNT(ename) cnt
FROM (
SELECT ename, sal, CASE
WHEN sal < 1500 THEN 'POOR'
```

```

WHEN sal BETWEEN 1500 AND 2500 THEN 'MID'
ELSE 'RICH'
END cat
FROM emp
) AS ec
GROUP BY cat;

```

- Find emp with highest sal in each dept.

```

SELECT deptno, MAX(sal) maxsal FROM emp
GROUP BY deptno;

```

```

SELECT e.ename, e.sal, e.deptno
FROM (
SELECT deptno, MAX(sal) maxsal FROM emp
GROUP BY deptno
) AS dm
INNER JOIN emp e ON e.deptno = dm.deptno
WHERE e.sal = dm.maxsal;

```

CTE

```

WITH ec AS (
SELECT ename, sal, CASE
WHEN sal < 1500 THEN 'POOR'
WHEN sal BETWEEN 1500 AND 2500 THEN 'MID'
ELSE 'RICH'
END cat
FROM emp
)
SELECT cat, COUNT(ename) cnt FROM ec
GROUP BY cat;

```

```

WITH dm AS (
SELECT deptno, MAX(sal) maxsal FROM emp
GROUP BY deptno
)
SELECT e.ename, e.sal, e.deptno FROM dm
INNER JOIN emp e ON e.deptno = dm.deptno
WHERE e.sal = dm.maxsal;

```

Window Functions + CTE

- Find emp with highest sal in each dept.

```
SELECT deptno, ename, sal,  
RANK() OVER(w) rnk FROM emp  
WINDOW w AS (PARTITION BY deptno ORDER BY sal DESC);  
  
WITH cte AS (  
SELECT deptno, ename, sal,  
RANK() OVER(w) rnk FROM emp  
WINDOW w AS (PARTITION BY deptno ORDER BY sal DESC)  
)  
SELECT * FROM cte  
WHERE rnk = 1;
```

- Find emp with second highest sal in each dept.

```
WITH cte AS (  
SELECT deptno, ename, sal,  
DENSE_RANK() OVER(w) rnk FROM emp  
WINDOW w AS (PARTITION BY deptno ORDER BY sal DESC)  
)  
SELECT * FROM cte  
WHERE rnk = 2;
```

Assignments

- <https://leetcode.com/problems/rank-scores/>
- <https://leetcode.com/problems/second-highest-salary/>
- <https://leetcode.com/problems/nth-highest-salary>
- <https://leetcode.com/problems/department-highest-salary/description/>
- <https://leetcode.com/problems/department-top-three-salaries/>
- <https://leetcode.com/problems/rising-temperature/>
- <https://leetcode.com/problems/consecutive-numbers>